

RED LAKE, ON, Canada

WMO: 718540

Lat: 51.070N

Lon: 93.790W

Elev: 386

StdP: 96.77

Time Zone: -6.00 (NAC)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-31.1	-37.7	0.1	-33.4	-35.1	0.1	-30.9	8.9	-14.4	8.0	-13.9	1.9	250	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	28.2	18.8	26.5	18.2	25.0	17.1	20.8	25.5	19.6	23.9	18.6	22.9	4.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.7	23.0	18.2	13.7	21.5	17.1	12.8	20.5	62.1	25.7	57.9	23.7	54.4	22.8	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.4	6.5	-36.9	31.2	1.9	1.8	-38.3	32.5	-39.4	33.6	-40.5	34.6	-41.9	36.0		
			-37.1	23.0	1.9	1.9	-38.4	24.3	-39.5	25.4	-40.6	26.5	-41.9	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-17.5	-15.9	-7.1	0.9	9.9	15.6	17.9	17.1	10.9	3.4	-6.2	-14.7
	DBStd	14.04	8.05	7.31	7.39	6.33	5.57	3.78	3.19	3.50	4.77	4.67	6.55	8.27
	HDD10.0	3971	851	726	531	279	72	3	0	1	44	211	486	766
	HDD18.3	6308	1109	959	789	524	266	97	48	66	227	463	736	1024
	CDD10.0	791	0	0	0	5	69	172	244	222	72	7	0	0
	CDD18.3	87	0	0	0	0	5	16	33	28	4	0	0	0
	CDH23.3	747	0	0	0	0	77	149	259	237	25	0	0	0
	CDH26.7	138	0	0	0	0	14	24	49	48	3	0	0	0
(14) Wind	WSAvg	3.3	3.1	3.1	3.4	3.4	3.3	3.2	3.1	2.9	3.5	3.6	3.6	3.0
(15) Precipitation	PrecAvg	668	32	23	34	40	57	96	89	91	76	54	47	32
	PrecMax	856	82	72	81	118	112	174	165	163	148	191	104	76
	PrecMin	394	4	6	4	4	10	32	35	22	7	8	13	13
	PrecStd	104	19	13	19	31	27	35	32	35	33	39	27	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	5.6	10.1	20.5	28.7	29.2	30.4	30.2	26.6	19.0	12.8	3.0
		MCWB	1.2	2.0	4.9	11.9	16.7	17.9	20.3	19.4	19.3	13.2	9.3	0.7
	2%	DB	0.2	2.2	7.0	16.2	25.2	26.7	28.0	27.9	22.8	15.8	6.7	-0.3
		MCWB	-0.4	0.0	2.5	9.3	15.0	17.1	19.3	19.3	17.5	11.5	4.8	-1.2
	5%	DB	-3.0	-0.1	4.8	13.2	22.9	24.9	26.1	26.1	20.4	13.2	4.6	-1.7
		MCWB	-3.7	-1.5	1.6	6.6	13.8	16.4	18.2	18.6	15.4	9.8	3.4	-2.2
	10%	DB	-5.9	-4.2	2.9	10.9	20.0	23.2	24.5	24.0	18.5	10.7	1.9	-3.2
		MCWB	-6.6	-5.5	0.4	5.4	12.6	15.4	17.3	17.9	14.9	7.7	0.6	-3.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	2.9	5.3	13.1	19.3	20.4	22.5	22.1	20.7	14.4	9.0	0.7
		MCDB	2.1	5.2	8.8	18.9	24.0	25.0	27.3	26.5	24.7	18.0	12.5	2.7
	2%	WB	-0.2	0.3	3.3	9.7	17.3	18.7	20.9	20.8	18.5	12.1	5.6	-1.0
		MCDB	0.3	1.8	6.1	14.3	22.2	24.0	25.5	25.6	21.0	14.7	6.7	-0.4
	5%	WB	-3.6	-1.9	1.9	7.6	15.2	17.7	19.6	19.6	16.9	10.0	3.3	-2.2
		MCDB	-3.0	0.0	4.1	12.7	21.4	22.4	23.6	24.0	19.4	12.7	4.5	-1.7
	10%	WB	-6.6	-5.3	0.6	5.5	13.2	16.6	18.5	18.5	14.9	7.8	0.8	-3.6
		MCDB	-6.0	-4.3	2.6	10.5	19.4	21.1	22.7	22.9	17.7	10.8	1.8	-3.2
(35) Mean Daily Temperature Range	MDBR		9.6	11.5	12.2	11.7	12.0	11.4	10.4	10.8	9.1	7.1	6.6	7.9
	5% DB	MCDBR	9.4	11.8	11.5	15.2	15.7	14.3	13.1	13.3	12.1	11.6	7.8	6.2
		MCWBR	9.0	10.2	8.0	9.1	7.6	6.6	6.5	6.5	7.2	7.4	6.0	5.4
	5% WB	MCDBR	9.4	11.4	9.7	13.8	13.0	11.2	10.7	11.6	10.1	10.6	7.3	5.6
		MCWBR	9.0	9.9	7.0	9.0	7.1	6.2	6.3	6.7	6.9	7.2	5.9	4.9
(40) Clear-Sky Solar Irradiance	taub		0.248	0.266	0.286	0.323	0.342	0.362	0.386	0.359	0.336	0.293	0.275	0.240
	taud		2.293	2.261	2.332	2.419	2.467	2.435	2.382	2.473	2.528	2.602	2.444	2.338
	Ebn at Noon		793	871	921	920	913	894	863	874	859	848	760	749
	Edh at Noon		64	88	100	104	104	109	113	98	83	64	56	52
(44) All-Sky Solar Radiation	RadAvg		1.13	2.31	3.59	5.07	5.30	5.43	5.57	4.64	3.30	1.88	0.99	0.76
	RadStd		0.13	0.19	0.29	0.36	0.34	0.46	0.45	0.40	0.37	0.23	0.10	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page